

Accuracy of Intraset Repetitions-in-Reserve Predictions During the Bench Press Exercise in Resistance-Trained Male and Female Subjects

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Abstract

Refalo, MC, Remmert, JF, Pelland, JC, Robinson, ZP, Zourdos, MC, Hamilton, DL, Fyfe, JJ, and Helms, ER. Accuracy of intraset repetitions-in-reserve predictions during the bench press exercise in resistance-trained male and female subjects. *J Strength Cond Res* XX(X): 000–000, 2023—This study assessed the accuracy of intraset repetitions-in-reserve (RIR) predictions to provide evidence for the efficacy of RIR prescription as a set termination method to inform proximity to failure during resistance training (RT). Twenty-four resistance trained male ($n = 12$) and female ($n = 12$) subjects completed 2 experimental sessions involving 2 sets performed to momentary muscular failure (barbell bench press exercise) with 75% of 1 repetition maximum (1RM), whereby subjects verbally indicated when they perceived to have reached either 1 RIR or 3 RIR. The difference between the predicted RIR and the actual RIR was defined as the “RIR accuracy” and was quantified as both *raw* (i.e., direction of error) and *absolute* (i.e., magnitude of error) values. High raw and absolute mean RIR accuracy (-0.17 ± 1.00 and 0.65 ± 0.78 repetitions, respectively) for 1-RIR and 3-RIR predictions were observed (including all sets and sessions completed). We identified statistical equivalence (equivalence range of ± 1 repetition, thus no level of statistical significance was set) in raw and absolute RIR accuracy between (a) 1-RIR and 3-RIR predictions, (b) set 1 and set 2, and (c) session 1 and session 2. No evidence of a relationship was found between RIR accuracy and biological sex, years of RT experience, or relative bench press strength. Overall, resistance-trained individuals are capable of high absolute RIR accuracy when predicting 1 and 3 RIR on the barbell bench press exercise, with a minor tendency for underprediction. Thus, RIR prescriptions may be used in research and practice to inform the proximity to failure achieved upon set termination.

Key Words: resistance training, proximity to failure, repetitions in reserve

Introduction

Proximity to failure during resistance training (RT) can be quantified as the number of “repetitions in reserve” (RIR) before reaching momentary muscular failure (i.e., when completion of the concentric portion of a repetition is not possible with a full range of motion without deviation from the prescribed exercise form) (9,35). For example, a proximity to failure of 1 RIR indicates that a single additional repetition could be completed, whereas 0 RIR indicates that the next attempted repetition would result in momentary muscular failure. Importantly, the proximity to failure in which an RT set is terminated may influence long-term RT outcomes such as strength development (5) and muscle hypertrophy (28) along with the neuromuscular fatigue incurred from RT (27,33).

Despite the importance of terminating RT sets at a given RIR to promote the desired physiological adaptations, common methods

of RT prescription lead to uncertainties in the RIR achieved upon set termination (26). For example, some studies have used a predetermined repetition prescription with a given relative load (e.g., 3 sets of 5 repetitions with 75% of 1 repetition maximum [1RM]), but considering the within-individual and between-individual variability in the maximum number of repetitions possible with the same relative load (4,29), some individuals may reach momentary muscular failure, whereas others may have many RIR. Other studies (1,23,24) have also attempted to control set termination during RT by having an individual terminate a given set once a specific percentage of “velocity loss” from the first (or fastest) repetition has been reached. However, velocity loss is not inherently individualized, is both exercise specific and load specific, and the velocity loss that corresponds to a given RIR varies from set to set (25). In addition, despite the objectivity of this velocity-based method of set termination, recent data (13) have found that percentage of velocity loss does not accurately predict repetitions performed in a set; thus, it cannot accurately quantify RIR.

To rectify the limitations of set prescriptions based on repetition maximum loads and percentage of velocity loss, an individual may control the proximity to failure reached by performing repetitions with a given load until they perceive a given RIR target has been reached, known as self-reported prediction of RIR. However, the

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utility of RIR-based set prescription (e.g., 3 sets of 10–15 repetitions with 2-RIR) is contingent upon the accuracy of the individual's RIR prediction. A recent meta-regression found that individuals underpredict RIR, on average, by ~ 1 repetition (11), and other research has found that RIR predictions may improve when RT sets are performed closer to momentary muscular failure (2,20,34), with higher relative loads and a greater number of successive sets performed (10,15,35), and in resistance trained vs. untrained (31,34) individuals. However, considerable heterogeneity exists between studies assessing the accuracy of RIR predictions. Specifically, studies may use various definitions of “set failure” (i.e., the set termination criteria for the definition of “failure” used in a given study) when instructing individuals to predict RIR, which may render the true accuracy of the RIR prediction unclear (26). In other words, if an individual gives an RIR prediction on a set that is terminated at volitional failure (i.e., when an individual perceives they have reached the prescribed set termination criteria), it cannot be known what the true RIR would be if the individual was told to terminate the set at momentary muscular failure. It is also possible that some individuals confuse proximity to failure with perceptions of discomfort, leading to an underestimation of RIR (2).

Therefore, the purpose of this study was to assess the accuracy of intraset RIR predictions (1 and 3 RIR) on the barbell bench press exercise (75% of 1RM load) over 2 sessions in resistance-trained male and female subjects who terminated each set at momentary muscular failure and were familiarized with the difference between perceived discomfort and proximity to failure. We also explored the relationship between RIR accuracy and (a) years of RT experience, (b) biological sex, and (c) relative bench press strength. We hypothesized that RIR accuracy would be greater when subjects were required to predict a 1 RIR versus a 3 RIR and we expected RIR accuracy to be greater in the second experimental session versus the first.

Methods

Experimental Approach to the Problem

This was a randomized crossover trial whereby subjects attended 2 experimental sessions at a local gymnasium (at JPS Health & Fitness, Melbourne, Australia) approximately 48 hours apart. In the first experimental session, the 1RM load was determined for the flat barbell bench press and used to calculate the 75% of 1RM load for each subject. A repetitions-to-failure assessment was conducted for the bench press in both experimental sessions, involving 2 sets performed to momentary muscular failure. Subjects were required to verbally indicate when they believed they had reached a 1 or 3 RIR (in a randomized order) during the set, and

the difference between the predicted RIR and the actual RIR (i.e., the number of repetitions performed after the prediction was given until momentary muscular failure was reached) was recorded to assess RIR accuracy.

Subjects

A total of 12 male and 12 female subjects were recruited (Table 1). All subjects (a) were between 18 and 40 years old, (b) reported no existing musculoskeletal injuries or neuromuscular disorders, (c) confirmed they had not used anabolic steroids or any illegal agents known to increase muscle size for the previous year, and (d) had ≥ 3 years of RT experience with a minimum of ≥ 3 weekly RT sessions. The mean 1RM for the bench press exercise was also greater than 120 and 80% of body mass for male and female subjects, respectively, indicating an advanced RT status as specified by Santos Junior et al. (30). All subjects reported experience working with a private fitness coach in a face-to-face setting, 12 subjects (50%) declared that they had previously competed in strength or physique sports (e.g., powerlifting or bodybuilding), and 23 subjects (96%) reported experience with intraset RIR prediction. The study procedures were approved by the Deakin University Human Research Ethics Committee (reference number: 2021-407). Before the commencement of data collection, all subjects were informed of the benefits and risks of the investigation (through plain language statement) before signing an institutionally approved informed consent from.

Sample Size Justification. This study was based on a secondary analysis of another main study (27). The target sample size of the main study (24 subjects) was based on outcome measures other than RIR accuracy (e.g., neuromuscular fatigue) and the following pragmatic considerations: (a) recruiting more than 24 subjects was not feasible given resource constraints, including the time and costs associated with data collection and subsequent analyses and (b) the chosen sample size is greater than most published studies investigating the influence of RT proximity to failure on neuromuscular fatigue using similar research designs (6,19,21,22,24). As α -priori sample size calculation was not performed, we conducted a sensitivity power analysis using G*Power software (Version 3.1.9.7) and calculated the minimum (critical) effect size (Cohen's $d = 0.27$) that could be statistically rejected based on a prespecified sample size ($n = 24$) and both type I (0.05) and type II (0.20) error rates.

Procedures

The following protocols were part of the “pretesting” phase for an original research study designed to assess the influence of

Table 1
Baseline subject characteristics.*†

Variable	Male Subjects ($n = 12$)		Female Subjects ($n = 12$)	
	Mean $\pm$ SD	Range	Mean $\pm$ SD	Range
Age (y)	28.5 $\pm$ 5.3	19–39	31.6 $\pm$ 5.7	23–40
Body mass (kg)	85.1 $\pm$ 8.3	74–99	62.3 $\pm$ 11.0	52–90
RT experience (y)	8.3 $\pm$ 3.7	3–15	7.2 $\pm$ 2.3	4–13
RT frequency (per wk)	4.4 $\pm$ 0.7	3–5	4.4 $\pm$ 0.7	4–6
1RM BP (kg)	116.0 $\pm$ 20.8	92.5–157.5	54.9 $\pm$ 13.0	35–77.5
Relative 1RM BP strength (kg·kg ⁻¹ bodyweight)	1.37 $\pm$ 0.26	1.14–1.93	0.88 $\pm$ 0.17	1.56–1.08

*y = years; kg = kilograms; RT = resistance training; 1RM = 1 repetition maximum; BP = bench press.

†An overview of the relevant characteristics for each subject. Data shown are mean $\pm$ SD. Relative strength calculated as barbell bench press 1RM (kg) divided by bodyweight (kg).

proximity to failure on neuromuscular fatigue and are extensively described elsewhere (27).

Prestudy Familiarization. Approximately 1 month before the commencement of the study period, subjects underwent a pre-study familiarization to establish appropriate exercise technique. Subjects performed 2 sets of 5 repetitions with the minimum load on the BP exercise to ensure appropriate technique as follows: the advanced subjects used and replicated their own lifting grip based on their previous experience with the BP exercise (at minimum, the barbell had to be grasped slightly outside shoulder width) and lowered the barbell until it contacted their chest (below the nipple line) and then lifted it back to the starting position without excessive bouncing off the chest or raising of the shoulders, trunk, or glutes off the bench. Subjects were then familiarized with performing the concentric (lifting) phase of each repetition with maximal intended lifting velocity (i.e., as fast as possible), followed by a controlled eccentric (lowering) phase (~2 seconds). Once subjects were familiarized with this lifting strategy, they were told to incorporate the BP into their own RT regimen and continue practicing with maximal intended lifting velocity until the commencement of the study.

One-Repetition Maximum Load Assessment. Upon commencement of the study, and after refamiliarization with correct exercise technique, subjects completed a 1RM assessment for the BP. First, a warm-up consisting of 1 set of 5 repetitions was performed with the minimum possible load (20 kg). The load was then progressively increased (15- to 20-kg increments for male subjects and 5–10 kg for female subjects) until the lifting velocity was lower than $0.5 \text{ m} \cdot \text{s}^{-1}$. Thereafter, the load was increased in smaller increments (2.5–10 kg for men, and 1.25–5 kg for women) until the 1RM was determined, defined as the heaviest load with which a single repetition was possible with a full range of motion. For the lighter loads ($>1 \text{ m} \cdot \text{s}^{-1}$), 3 repetitions were performed at each load, 2 repetitions were performed for the moderate loads, and a single repetition for the heavier loads ($<0.5 \text{ m} \cdot \text{s}^{-1}$).

Experimental Conditions and Repetitions-to-Failure Assessments. Subjects completed 2 experimental sessions (48 hours apart) involving a repetitions-to-failure assessment. A standardized warm-up was first completed, starting with 20 kg, then 50, 65, and 85% of the 75% of 1RM load (for 6, 5, 4 and 3 repetitions, respectively, with 2-minute interset rest periods). Subjects were then briefed about intraset RIR prediction, and it was made clear that 0 RIR indicates the last full range-of-motion repetition possible before momentary muscular failure is reached (i.e., if a subsequent repetition was attempted, momentary muscular failure would occur). Momentary muscular failure was defined as the point where despite attempting to do so, the individual was unable to complete the concentric portion of their current repetition with a full range of motion without deviation from the prescribed form of the exercise. Considering the possibility for subjects to conflate RIR predictions with perceptions of discomfort (31), subjects were also briefed about the difference between perceived discomfort and subjective perception of proximity to failure. Two sets to momentary muscular failure with the load corresponding to 75% of 1RM were then performed. In a randomized order, subjects verbally indicated when they believed that they had reached 3 RIR during 1 of the sets and then indicated when they believed that they had reached a 1 RIR during the other set, before continuing to perform repetitions to momentary muscular failure.

Objective Outcome Measures. The difference between the predicted RIR (i.e., 1- or 3-RIR) and the actual RIR (i.e., number of repetitions performed after the prediction was given until momentary muscular failure was reached) was defined as the RIR accuracy. This was calculated as both the *raw* RIR accuracy, which accounts for directionality of error (i.e., negative values indicate underestimation of RIR, and positive values indicate overestimation of RIR), and the *absolute* RIR accuracy, which is an absolute value (i.e., difference in predicted vs. actual RIR regardless of directionality) that represents the magnitude of error. For example, if a subject gave their 3-RIR prediction and completed 5 more repetitions, the raw RIR accuracy would be -2 repetitions (indicating the subject underestimated their capabilities by 2 repetitions) and the absolute RIR accuracy would be calculated as the absolute value of the raw RIR accuracy (i.e., 2 repetitions).

Statistical Analyses

Primary Outcomes. All statistical analyses were performed using “R” software (v 4.0.2; R Core Team, <https://www.r-project.org/>). In answering our research questions, we opted to avoid dichotomizing our findings and therefore did not use traditional null hypothesis significance testing (12,18). Instead, we considered the outcomes based on the magnitude of effect estimates and associated confidence interval (CI) width, with greatest emphasis placed on the precision of point estimates. To investigate the accuracy of intraset RIR predictions, separate mixed-effects models were generated for raw and absolute RIR accuracy, respectively. Raw RIR accuracy was analyzed with a linear mixed-effects model using restricted maximum likelihood estimation (REML). However, absolute RIR accuracy was analyzed with a generalized, linear, mixed-effects model with a Poisson’s error distribution (specified with a log-link function and REML because of positively skewed count data of the dependent variable). Both models included fixed effects for (a) the RIR target at which accuracy was evaluated, (b) the set number within each session, (c) the order of the experimental sessions, and (d) relevant interactions thereof (i.e., session $\times$ set, session $\times$ RIR, set $\times$ RIR). Random intercepts were included per subject to account for repeated measures, and each model was adjusted for the number of repetitions performed per set, which was included as a covariate.

To address the primary research questions, estimated marginal means and their comparisons were produced with both 90 and 95% CIs within each fixed effect after adjusting for the other independent variables (i.e., session, set, and RIR) and the number of repetitions performed per set. Statistical equivalence was evaluated for each comparison, including (a) 1-RIR and 3-RIR conditions, (b) first set and second set, and (c) first session and second session (14). Specifically, if the bounds of the CIs for a given comparison are within the equivalence range set at the smallest effect size of interest, RIR accuracy is considered statistically equivalent (i.e., no meaningful difference). The smallest effect size of interest was set at ± 1 repetition (i.e., equivalence range) because recent meta-analytic data suggested that a 1-repetition difference in proximity to failure is unlikely to influence long-term muscle hypertrophy outcomes with RT (28). Furthermore, the upper and lower bounds of the equivalence range, and their overlap with the estimated effect, were used to evaluate each comparison. Finally, interactions of fixed effects were evaluated visually and through model regression coefficients with their respective 95% CIs. All estimated marginal means from the Poisson’s model were back transformed to the response scale before marginal averaging using the “emmeans” package to allow for additive

comparisons. Raw study data, model outputs, and visualizations are presented in the Supplemental Digital Content 1 (see Supplementary Materials, <http://links.lww.com/JSCR/A433>).

Secondary/Exploratory Outcomes. To explore the influence of RT experience (years), biological sex, and relative bench press strength on RIR prediction accuracy, multiple linear regression models were fit separately for raw and absolute RIR accuracy. Subject scores were averaged so that only 1 observation existed per subject for raw and absolute RIR accuracy, respectively. To eliminate unnecessary collinearity of model predictors, a correlation matrix was constructed with all candidate model predictors ($n = 4$) and only the strongest predictors were considered for further exploration ($r = >0.20$ with dependent variable). For raw RIR accuracy, the predictors included in the final models were biological sex and years of RT experience. For absolute RIR accuracy, the predictors included in the final model were biological sex and relative bench press strength. For both exploratory models, the reference level of biological sex was set to female.

Results

Overall, subjects recorded a *raw* RIR accuracy of -0.23 ± 0.75 repetitions (range = 1 to -3 repetitions) for 1-RIR predictions and -0.10 ± 1.21 repetitions (range = 2 to -3 repetitions) for 3-RIR predictions, with a combined (1- and 3-RIR) raw RIR accuracy of -0.17 ± 1.00 repetitions (range = 2 to -3 repetitions; Figure 1A). Furthermore, an *absolute* RIR accuracy of 0.40 ± 0.68 repetitions (range = 0-3 repetitions) was recorded for 1-RIR predictions and 0.90 ± 0.81 repetitions (range = 0-3 repetitions) for 3-RIR predictions, with 0.65 ± 0.78 repetitions (range = 0-3 repetitions) for 1-RIR and 3-RIR predictions combined (Figure 1B). Individual raw and absolute RIR accuracy for each subject is displayed in Figure 2.

Raw Repetitions-in-Reserve Accuracy

Raw RIR accuracy was comparable between RIR predictions made at 1 RIR (-0.23 repetitions [95% CI: $-0.52, 0.06$]) and 3 RIR (-0.15 repetitions [95% CI: $-0.45, 0.14$]) and between set 1 (-0.13 repetitions [95% CI: $-0.43, 0.17$]) and set 2 (-0.25 repetitions [95% CI: $-0.56, 0.05$]). The 95% CIs of the comparisons in RIR accuracy between 1 RIR and 3 RIR (-0.08 repetitions [95% CI: $-0.49, 0.33$]) and between set 1 and set 2 (0.12 repetitions [95% CI: $-0.31, 0.56$]) indicated that these data are compatible with a null effect (i.e., no meaningful differences between “RIR condition” and “set number” were found). Conversely, slightly greater RIR underprediction was found in session 2 (-0.44 repetitions [95% CI: $-0.73, -0.16$]) compared with session 1 (0.06 repetitions [95% CI: $-0.24, 0.36$]) because the 95% CI of the comparison in RIR accuracy between session 1 and session 2 did not contain the null-point estimate (0.51 repetitions [95% CI: $0.09, 0.92$]).

When evaluating statistical equivalence with an equivalence range of ± 1 repetition, raw RIR accuracy was considered statistically equivalent for all comparisons (Figure 3A); 1 RIR vs. 3 RIR (-0.08 repetitions [90% CI: $-0.42, 0.26$]), set 1 vs. set 2 (0.12 repetitions [90% CI: $-0.24, 0.49$]), and session 1 vs. session 2 (0.51 repetitions [90% CI: $0.16, 0.85$]). Furthermore, we found no evidence of interactions between variables (i.e., RIR, set, and session; see Figure 2.3, Supplemental Digital Content 2, <http://links.lww.com/JSCR/A434>); the 95% CIs for all model regression coefficients indicated that the data were compatible with a null

effect and can be found within Supplemental Digital Content 1 (see Table 1.1, <http://links.lww.com/JSCR/A433>).

Absolute Repetitions-in-Reserve Accuracy

Absolute RIR accuracy was comparable between RIR predictions made in session 1 (0.72 repetitions [95% CI: $0.42, 1.02$]) and session 2 (0.60 repetitions [95% CI: $0.35, 0.86$]) and between set 1 (0.60 repetitions [95% CI: $0.33, 0.86$]) and set 2 (0.73 repetitions [95% CI: $0.43, 1.03$]). The 95% CIs of the comparisons in RIR accuracy between session 1 and session 2 (0.11 repetitions [95% CI: $-0.24, 0.47$]) and between set 1 and set 2 (-0.13 repetitions [95% CI: $-0.50, 0.24$]) indicated that the data were compatible with a null effect (i.e., no meaningful differences between “session number” and “set number” were found). Conversely, slightly less RIR prediction error was found for RIR predictions made at 1 RIR (0.39 repetitions [95% CI: $0.19, 0.59$]) compared with 3 RIR (0.93 repetitions [95% CI: $0.59, 1.28$]) because the 95% CI of the comparison in RIR accuracy between 1 RIR and 3 RIR did not contain the null-point estimate (-0.54 repetitions [95% CI: $-0.91, -0.18$]).

When evaluating statistical equivalence with an equivalence range of ± 1 repetition, absolute RIR accuracy was considered statistically equivalent for all comparisons (Figure 3B); 1 RIR vs. 3 RIR (-0.54 repetitions [90% CI: $-0.85, -0.24$]), set 1 vs. set 2 (-0.13 repetitions [90% CI: $-0.44, 0.18$]), and session 1 vs. session 2 (0.11 repetitions [90% CI: $-0.18, 0.41$]). Furthermore, we found 1 potential interaction between session and set (see Figure 2.3, Supplemental Digital Content 2, <http://links.lww.com/JSCR/A434>), and the 95% CI of the model regression coefficient did not contain the null-point estimate ($\beta = 4.24$ [95% CI: $1.38, 13.02$]), suggesting that RIR prediction error (for both 1- and 3-RIR predictions, combined) decreased in session 1 from set 1 (0.87 repetitions [95% CI: $0.43, 1.31$]) to set 2 (0.57 repetitions [95% CI: $0.21, 0.93$]), but in session 2, it increased from set 1 (0.32 repetitions [95% CI: $0.08, 0.57$]) to set 2 (0.89 repetitions [95% CI: $0.45, 1.32$]). The remaining 95% CIs for all model regression coefficients were compatible with a null effect and can be found within Supplemental Digital Content 1 (see Table 1.2, <http://links.lww.com/JSCR/A433>).

Exploratory Predictors of Raw and Absolute Repetitions-in-Reserve Accuracy

The multiple regression model for raw [$R^2 = 0.22$, adjusted $R^2 = 0.15$] and absolute [$R^2 = 0.08$, adjusted $R^2 = 0.00$] RIR accuracy demonstrated weak predictive capacity [$R^2 = 0.22$, adjusted $R^2 = 0.15$]. All visuals and summary outputs of the exploratory models can be found in the Supplemental Digital Content 1 (see Supplementary File S3, <http://links.lww.com/JSCR/A433>).

Discussion

The purpose of this study was to examine the accuracy of intraset RIR predictions (1- and 3-RIR) across multiple sets and sessions on the barbell bench press exercise in resistance-trained subjects. High *raw* and *absolute* RIR accuracy (-0.17 ± 1.00 and 0.65 ± 0.78 repetitions, respectively) was observed (Figure 1) and based on our smallest effect size of interest (i.e., equivalence range of ± 1 repetition), we conclude statistically equivalent RIR accuracy between (a) 1- and 3-RIR predictions, (b) set 1 and set 2, and (c) session 1 and session 2 (Figure 3). Furthermore, our exploratory analyses found a negligible impact of the variables of interest (i.e., years of RT

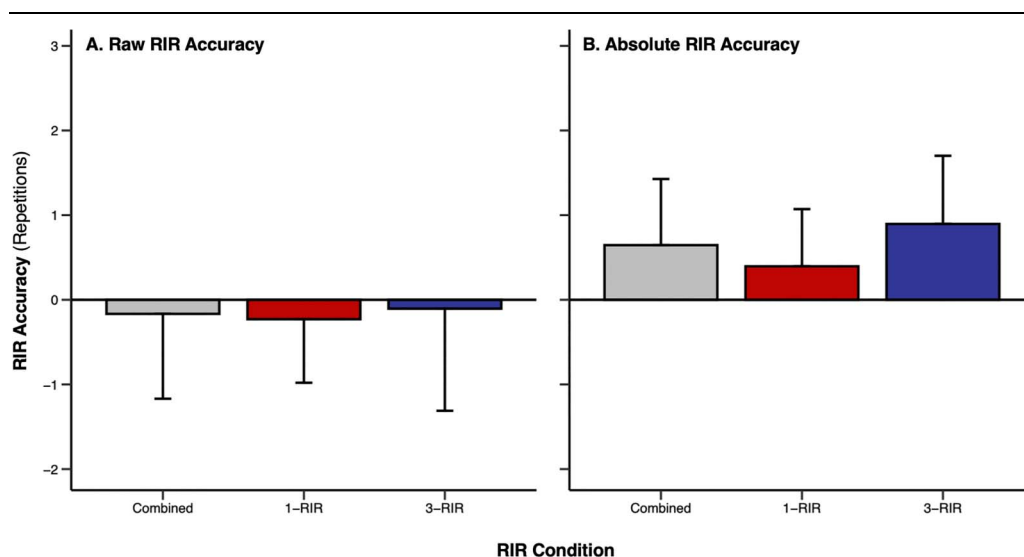


Figure 1. Raw (A) and absolute (B) mean RIR accuracy for combined (1 and 3 RIR) and individual 1-RIR and 3-RIR predictions. Data shown are presented as mean $\pm$ SD, averaged across all sets and sessions performed. Zero (on the y-axis) indicates accurate RIR predictions. Individual subject values are displayed in Figure 2 to highlight interindividual variability in RIR accuracy. RIR, repetitions in reserve.

experience, biological sex, and relative bench press strength) on RIR accuracy. Both hypotheses (i.e., RIR accuracy would be greater when subjects were required to predict a 1 RIR vs. a 3 RIR and would be greater in session 2 compared with session 1) were thus not supported by our findings, likely because of the high RIR accuracy that subjects achieved in each experimental session.

Our primary finding suggests that resistance-trained individuals can demonstrate high RIR accuracy on the barbell bench press exercise when instructed to subjectively predict 1 or 3 RIR during a set performed to momentary muscular failure. To our knowledge, we recruited 1 of the most highly resistance-trained samples of subjects examined within the relevant literature (11) (RT experience: men = 8.3 ± 3.7 and women = 7.2 ± 2.3 years; relative bench press strength: men = 1.37 ± 0.26 and women = 0.88 ± 0.17 kg·kg⁻¹ body mass), of which, 96% had previous experience using RIR as a set termination method. Although it is likely that the extensive RT experience of our subjects contributed to the high *absolute* RIR accuracy observed (0.65 ± 0.78 repetitions), with a minor trend for

underprediction (raw RIR accuracy = -0.17 ± 1.00 repetitions), these findings are in line with meta-analytic data revealing that subjects of *any* RT experience across 13 studies also achieved an absolute RIR accuracy of less than 1 repetition (0.95 repetitions [CI: 0.17–1.73]) (11). Also likely influencing the high absolute RIR accuracy observed in this study was our effort to brief subjects about the difference between *perceived discomfort* (i.e., unpleasant sensations perceived during exercise (16)) and proximity to failure. Perceived discomfort increases as proximity to failure nears (26) because of multiple factors including elevated metabolite accumulation, breathing rate, and body temperature (17), ultimately increasing local pain perception (through group III/IV muscle afferent activation (17)) and requiring greater cognitive effort to complete further repetitions (3). Although afferent feedback does not seem to contribute substantially to perception of effort during exercise (16), making it possible to differentiate between perceived discomfort and proximity to failure (i.e., perceived effort), it is likely that RIR predictions in previous research have been influenced by perceived

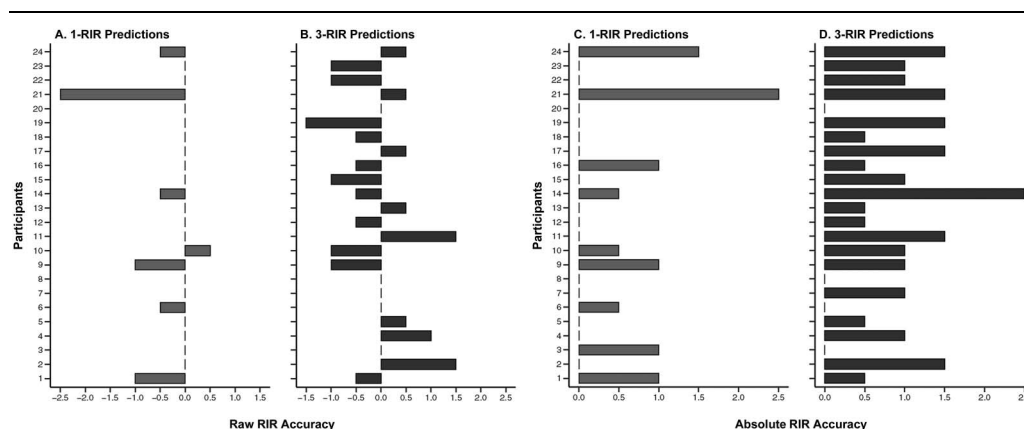


Figure 2. Individual subject values for the mean raw and absolute RIR accuracy of (A,C) 1-RIR, and (B,D) 3-RIR predictions. Data shown are presented as mean $\pm$ SD, averaged across all sets and sessions performed. RIR accuracy corresponding to zero (and shaded) indicates accurate RIR predictions. Positive RIR accuracy values indicate overprediction, and negative values indicate underprediction. RIR, repetitions in reserve.

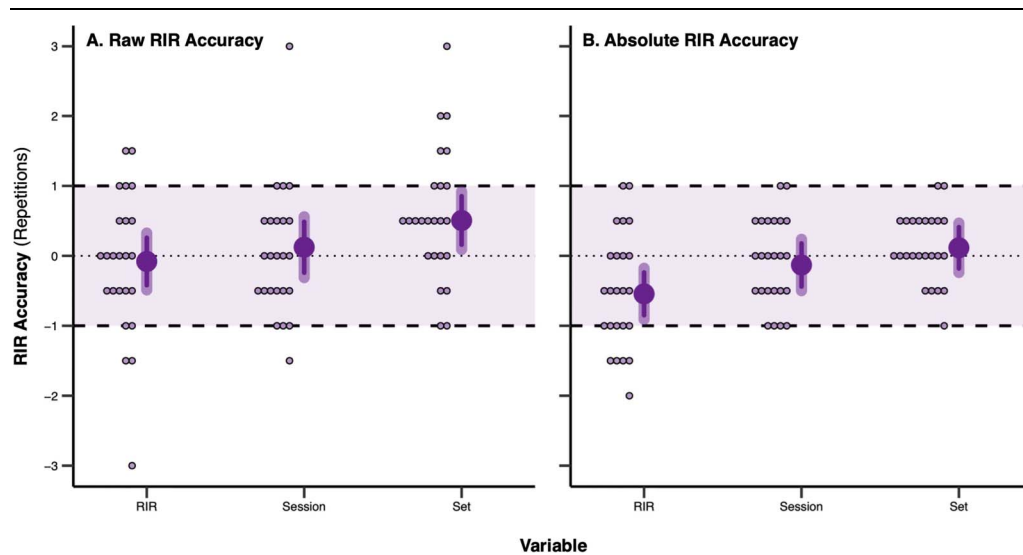


Figure 3. Estimated marginal mean values with equivalence range showing difference in raw (A) and absolute (B) RIR accuracy between RIR predictions (1 and 3 RIR), sessions (1 and 2), and sets (1 and 2). Point estimate presented as estimated marginal mean with 90% (darker colored bands) and 95% CIs (lighter colored bands). Smallest effect size of interest was set at ± 1 repetition, generating an equivalence range (indicated by the shaded area) with a lower equivalence bound of -1 repetition and upper equivalence bound of 1 repetition (indicated by the dotted lines). Neither of the point estimates (or their CIs) crossed the upper or lower equivalence bounds, demonstrating statistical equivalence between 1 and 3 RIR, session 1 and session 2, and set 1 and set 2. Individual subject values also displayed to highlight interindividual variability in RIR accuracy for each comparison. RIR, repetitions in reserve.

discomfort (31,32). For example, one may confuse a given level of perceived discomfort with a specific proximity to failure, leading to erroneous RIR predictions that are based on individual tolerance to discomfort and not on perceptions of proximity to failure. This possibility highlights the importance of distinguishing between perceived discomfort and proximity to failure when using RIR prediction as a set termination method in research and practice. Overall, the RIR accuracy that should be deemed acceptable to use RIR in RT prescription remains unclear, as such, future research should continue exploring RIR accuracy and its implications for muscle hypertrophy by using RIR as a set termination method and employing extensive familiarization (e.g., performing RT to momentary muscular failure, practicing intraset RIR prediction, briefing on perceived discomfort).

Contrary to our findings that demonstrate similar accuracy between 1-RIR and 3-RIR predictions (Figure 2), previous research found that RIR accuracy improves when predictions are made closer to momentary muscular failure (2,20,34). For example, one study (34) found that when subjects verbally indicated a perceived 5, 3, and 1 RIR during a set to volitional failure at 70% 1RM on the barbell back squat exercise, RIR accuracy was greater as proximity to failure neared; specifically, RIR predictions were off by 5.15 ± 2.92 , 3.65 ± 2.46 , and 2.05 ± 1.73 repetitions, for 5, 3, and 1 RIR, respectively. Importantly, however, the exercise performed may influence the accuracy of RIR predictions (e.g., RIR predictions may be more accurate for upper-body vs. lower-body exercises (8)), possibly contributing to differences or lack thereof in RIR accuracy between RIR conditions across studies. The possibility also remains that meaningful differences in RIR accuracy between RIR conditions would have been evident if we had included RIR predictions even further from momentary muscular failure (e.g., 5 RIR). We also investigated differences in RIR accuracy between the first and second set performed and the first and second experimental session completed and found that neither variable (i.e., set and session number) meaningfully influenced RIR accuracy (Figure 2). These

findings are in contrast with previous research, suggesting that the accuracy of RIR predictions increases as successive sets are performed (7,10), likely because of the lowered number of repetitions performed as a result of neuromuscular fatigue. However, unlike our study, whereby subjects were instructed to verbally indicate when they had reached the RIR target at any point during each set, previous studies (7,10) required subjects to predict RIR at the same point within each set (e.g., following the 10th repetition), ultimately prompting RIR predictions closer to momentary muscular failure because of neuromuscular fatigue and, as such, improved RIR accuracy. Overall, our findings provide evidence that (a) resistance-trained individuals (with experience predicting RIR) can predict 1 RIR and 3 RIR with similar accuracy based on the equivalence range we deemed meaningful and (b) improvements in RIR accuracy across successive sets and sessions are likely dependent on the initial RIR accuracy (i.e., lower RIR accuracy allows for greater improvements) and may not be observed in resistance-trained individuals.

Our exploratory multiple regression analysis found no evidence of a relationship between (a) years of RT experience, (b) biological sex, or (c) relative bench press strength and RIR accuracy, likely because of the consistently high RIR accuracy that we observed; but considering the lack of previous research exploring the influence of these variables on RIR accuracy, future research is required to consolidate our findings.

The consistently high RIR accuracy of our resistance-trained subjects likely influenced the statistically equivalent RIR accuracy detected between 1 and 3 RIR predictions, set 1 and set 2, and session 1 and session 2. Furthermore, we only examined RIR accuracy on the barbell bench press exercise and whether our results can be generalized to other exercises or muscle groups is unclear and thus requires further research to elucidate. The qualified supervisors in this study did not count the repetitions performed by the subjects aloud during the experimental sessions and did not report the number of repetitions performed to the

subjects; but whether the subjects kept track of the number of repetitions performed is unknown and whether our findings apply when repetitions are counted and pacing strategies are implemented (as is common in practice) is unclear.

Our main findings show that resistance-trained individuals can demonstrate high *absolute* RIR accuracy when predicting 1 and 3 RIR (0.65 ± 0.78 repetitions), with a minor trend for underprediction indicated by the negative *raw* RIR accuracy observed (-0.17 ± 1.00 repetitions), on the barbell bench press exercise. Furthermore, likely because of the consistently high RIR accuracy of our subjects, we found statistically equivalent (within an equivalence range of ± 1 repetition) *raw* and *absolute* RIR accuracy between (a) 1 and 3 RIR predictions, (b) set 1 and set 2, and (c) session 1 and session 2. Overall, these findings provide evidence that RIR can be consistently predicted within 1 repetition of the RIR target, which we deemed meaningful and practically important.

Practical Applications

Considering the high absolute RIR accuracy observed in this study (Figure 1), when working with resistance-trained individuals, practitioners may choose to allocate a specific number of sets to a given exercise along with a repetition range and an RIR target, whereby individuals self-select a load that would prompt set termination at the perceived RIR within the prescribed repetition range. Considering that the proximity to failure reached during RT may influence the RT stimulus achieved (e.g., mechanical tension), using an RIR prescription to inform set termination may be advantageous when compared with a predetermined repetition prescription (e.g., 3 sets of 10 repetitions), whereby the proximity to failure reached upon set termination is unclear, and sets may be terminated with a potentially suboptimal proximity to failure. Indeed, an RIR prescription may inform the RT stimulus achieved within a given set relative to the maximum RT stimulus possible (e.g., if a set is terminated at 2 RIR on the 10th repetition, the maximum possible RT stimulus would have been reached on the 12th repetition), and the level of neuromuscular fatigue incurred (i.e., as proximity to failure nears, neuromuscular fatigue increases (27)). However, an RIR prescription may only be effectively executed with a certain level of RIR prediction accuracy (e.g., within ± 1 repetition of the RIR target) and may be most suitable to resistance-trained individuals. Thus, we suggest that individuals aiming to achieve maximal RIR prediction accuracy (a) learn appropriate and safe exercise technique that can be sustained during a set to momentary muscular failure, (b) understand the difference between perceived discomfort and proximity to failure to reduce the chance of inaccurate RIR predictions, and (c) practice reaching momentary muscular failure where safe to do so to improve perceptions of proximity to failure and “anchor” RIR targets (e.g., if an individual performs 12 repetitions to momentary muscular failure, a 2 RIR is anchored to the perception of the 10th repetition), before using an RIR prescription to inform set termination. Finally, to assess whether experience with RIR predictions improves RIR accuracy overtime, individuals should periodically test the accuracy of their RIR predictions when it is deemed safe to reach momentary muscular failure.

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